

PRAKTIKUM SISTEM OPERASI

MODUL 4

(Membaca_Source_code_xinu)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL/TP

1. Soal 1

Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

Jawaban:

Nama image hasil kompilasi Xinu:

xinu.elf

Ukuran file:

156 KB

Lokasi file:

~/xinu/compile

2. Soal 2

Membaca Source Code xinu

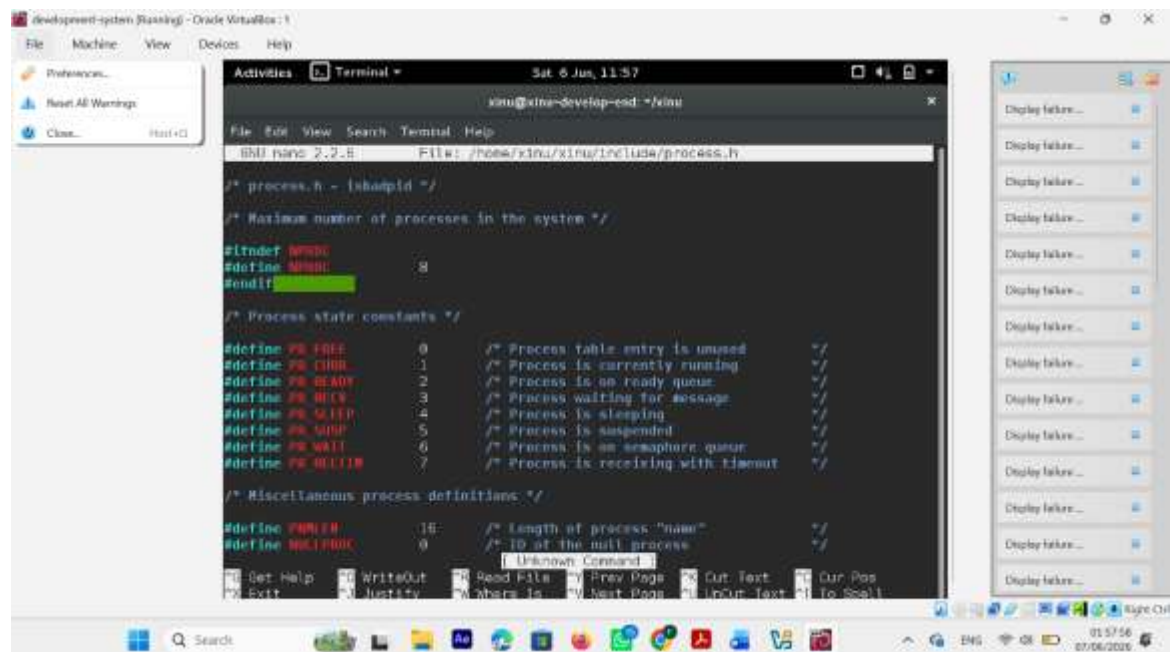
Jawaban:

Dilakukan eksplorasi source code Xinu yang terdiri dari direktori compile, config, device, include, lib, net, shell, dan system untuk memahami struktur sistem operasi Xinu.

3. Soal 3

Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut? Hint: file berekstensi .h

Jawaban:



```
development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal
xinu@xinu-develop-ost: ~/xinu
File Edit View Search Terminal Help
shl nano 2.2.8 File: /home/xinu/xinu/include/process.h

/* process.h - lshadpid */
/* Maximum number of processes in the system */
#define NPROC 8

/* Process state constants */
#define PS_FREE 0 /* Process table entry is unused */
#define PS_RUNNING 1 /* Process is currently running */
#define PS_READY 2 /* Process is on ready queue */
#define PS_WAITING 3 /* Process waiting for message */
#define PS_SLEEP 4 /* Process is sleeping */
#define PS_SUSPENDED 5 /* Process is suspended */
#define PS_WAITING_SEMAPHORE 6 /* Process is on semaphore queue */
#define PS_RECEIVING_TIMEOUT 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */
#define PROC_NAME_LEN 15 /* length of process "name" */
#define PROC_ID 0 /* ID of the null process */

Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Utility Where Is Next Page Up/Down Text To Shell
```

Struktur data proses pada sistem operasi Xinu terdapat pada file `xinu/include/process.h`. Struktur tersebut bernama `struct procent` dan digunakan untuk menyimpan informasi yang berkaitan dengan setiap proses yang berjalan pada sistem.

Informasi yang disimpan dalam struktur data proses antara lain:

- Process ID (PID), yaitu identitas unik setiap proses.
- Process State, yaitu status proses seperti `running`, `ready`, `sleep`, `suspend`, dan lain-lain.
- Process Priority, yaitu prioritas proses dalam penjadwalan CPU.
- Stack Pointer, yaitu alamat stack yang sedang digunakan proses.
- Stack Base dan Stack Length, yaitu alamat awal dan ukuran stack proses.
- Process Name, yaitu nama proses.
- Parent Process ID, yaitu ID proses induk yang membuat proses tersebut.
- Semaphore, yaitu semaphore yang sedang digunakan proses.
- Message dan Message Status, yaitu data pesan dan status penerimaan pesan antar proses.

Struktur data proses ini digunakan oleh kernel Xinu untuk melakukan manajemen proses dan penjadwalan CPU.

4. Soal 4

Mengubah welcome banner pada Xinu

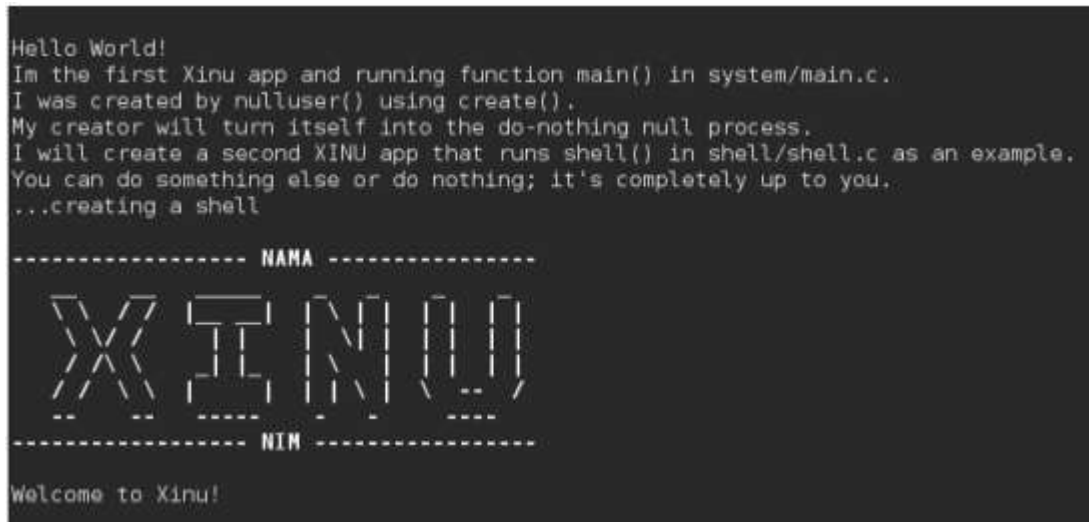
a. Carilah file yang menyimpan banner Xinu!

Hint: file berekstensi `.h` pada direktori `xinu/include`

b. Carilah file yang menampilkan banner Xinu!

Hint: file berekstensi `.c` pada direktori `xinu/shell`

C, Ubahlah welcome banner Xinu sehingga menjadi seperti ini:



Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda

d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux

- `$ cd xinu/compile`
- `$ make clean`
- `$ make`

- \$ sudo minicom
- Jalankan vm backend

Jawaban:

A.

```

development-system [running] - Oracle VM VirtualBox - 1
File Machine View Devices Help
Activities Terminal
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: /home/xinu/xinu/include/shell.h

/* Shell banner (assumes VT100) */
#define SHEL1 0x0001  "X\nM\n"
#define SHEL1 0x0002  "X\nM\n"
#define SHEL1 0x0003  "X\nM\n"
#define SHEL1 0x0004  "X\nM\n"
#define SHEL1 0x0005  "X\nM\n"
#define SHEL1 0x0006  "X\nM\n"
#define SHEL1 0x0007  "X\nM\n"
#define SHEL1 0x0008  "X\nM\n"
#define SHEL1 0x0009  "X\nM\n"
#define SHEL1 0x000A  "X\nM\n"
/* Messages shell displays for user */
#define SHEL1 0x000B  "xsh $ " /* command prompt */
#define SHEL1 0x000C  "Welcome to xinu!\n" /* Welcome message */
#define SHEL1 0x000D  "Shell closed\n" /* Shell exit message */
#define SHEL1 0x000E  "Syntax error\n" /* Syntax error message */
#define SHEL1 0x000F  "Cannot create process\n" /* command error */
#define SHEL1 0x0010  "Cannot open file %s for input\n" /* input err */
#define SHEL1 0x0011  "Cannot open file %s for output\n" /* output err */

Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Justify Where Is Next Page InOut Text To Spell

```

B.

```

development-system [running] - Oracle VM VirtualBox - 1
File Machine View Devices Help
Activities Terminal
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: /home/xinu/xinu/shell/shell.c

/* In background */
pid32_t tappid = segtab(nsegs-1).spid; /* last seg. pid */

if (! backgnd) {
    msg = receive();
    while (msg != tappid) {
        msg = receive();
    }
}

/* terminate the shell process by returning from the top level */
fprintf(dev, SHEL1, "X\nM\n");
return 0;
}

```

C.

The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu". Inside, the GNU nano 2.2.6 editor is open, editing the file "/home/xinu/xinu/include/shell.h". The file content defines shell banners and messages:

```
/* Shell banner (assumes VT100) */

#define SHELL_BAN0      "\033[31;1m"
#define SHELL_BAN1      "-----FarhanKurniawan-----"
#define SHELL_BAN2      "                                                                "
#define SHELL_BAN3      "          \_/_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN4      "         /_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN5      "        /_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN6      "       /_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN7      "      /_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN8      "     /_/_/_/_/_/_\_/_/_/_\_/_/_\_/_/_\_/"
#define SHELL_BAN9      "-----2311104073-----"
                        "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT    "xsh $ "           /* Command prompt                */
#define SHELL_STRMSG    "Welcome to XINU!\n" /* Welcome message                  */
#define SHELL_EXITMSG   "Shell closed\n"    /* Shell exit message               */
#define SHELL_SYNERMSG  "Syntax error\n"    /* Syntax error message             */
#define SHELL_CREATEMSG "Cannot create process\n" /* command error                   */
#define SHELL_INERRMSG  "Cannot open file %s for input\n" /* Input err                       */
#define SHELL_OUTERRMSG "Cannot open file %s for output\n" /* Output err                      */
```

At the bottom of the terminal, there are keyboard shortcuts for various actions:

Get Help	WriteOut	Read File	Prev Page	Cut Text	Cur Pos
Exit	Justify	Where Is	Next Page	UnCut Text	To Spell

```
Activities Terminal Sat 6 Jun, 13:32
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
115465 bytes of Xinu code.
[0x00100000 to 0x0011C308]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)

-----FarhanKurniawan-----
XINU
-----2311104073-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0
```